

Probabilistic Parameter Estimation of the US 10-Year Treasury Yield: A Frequentist and Bayesian Comparative Analysis for ASX Energy Stock Inference

COMP3018 – Probabilistic Machine Learning

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Artefact 1

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Table of Contents

Table of Acronyms	2
Capstone and Team Integration	3
Project Proposal	3
Problem Statement	3
Problem Objectives	3
Justification.....	3
Model Survey	5
Human Factors and Ethics Review.....	5
Stakeholder Analysis	5
Dataset Analysis	6
Source	6
Exploratory Data Analysis	6
Frequentist and Bayesian Comparative Analysis	7
System Architecture and Implementation	8
Design.....	8
Process	9
Technical Decisions.....	9
Appendix	9
Citations	9
AI Declaration.....	10

Table of Acronyms

Abbreviations	Expanded
ML	Machine Learning
DAG	Directed Acyclic Graph
MLE	Maximum Likelihood Estimation
MCMC	Markov Chain Monte Carlo
NUTS	No-U-Turn Sampler
VI	Variational Inference
KS	Kolmogorov-Smirnov
SW	Shapiro-Wilks
ETL	Extract, Transform, Load

Capstone and Team Integration

This study forms part of a collaborative probabilistic inference engine designed to estimate conditional probability of ASX energy stock price movements, using a Directed Acyclic Graph of causal feature nodes. Each group member will develop a data pipeline for ingestion and transformation for their causal feature, as well as the probabilistic parameter estimation. The individual node featured in this study is the US 10-Year Treasury Yield.

With the group theoretical DAG, the US 10-Year Treasury Yield acts as a parent node to the VIX and AUD/USD nodes. Rising yields signal monetary tightening conditions, increasing market fear captured by VIX and strengthening the USD which suppresses the AUD/USD exchange rate[1][2].

The individual study will output a standardised data contract containing three features; yield_pct, daily_change and vol_21d. The Bayesian posterior distributions μ , σ , and v are produced, providing full uncertainty quantification rather than point estimates. The vol_21d captures time-varying volatility clustering, allowing downstream modules to condition on periods of elevated uncertainty. Forward-filling of US public holidays ensures alignment with ASX trading calendar, crucial in maintain data governance integrity across the team. In Artefact2, yield_pct will serve as the state variable and daily_change as the emission variable for the Hidden Markov Model component.

Project Proposal

Problem Statement

ASX Energy stock prices are driven by a range of uncertain factors including heightened geopolitical tensions, investor hedging, prices of commodities, interest rates and government policy and regulation[3]. Standard machine learning (ML) models produces point predictions without quantifying uncertainty. This study will look into building a probabilistic engine to estimate the probability of energy stock price movements using a Directed Acyclic Graph (DAG) of ten causal features.

Problem Objectives

Objectives for this project are:

- To quantify parameter uncertainty in feature distributions by comparing Bayesian and Frequentist estimation methods.
- Build individual probabilistic sub-systems that will integrate into a group DAG.
- Produce outputs aware of uncertainty that will propagate through the inference engine.

Justification

Quantifying uncertainty is critical for this problem. Energy markets are inherently non-stationary; the parameters governing price behaviour shift across different macroeconomic regimes. This is evident in the US-10 Year Treasury Yield data used in this study. The yield transitioned from near-zero levels during the COVID pandemic to nearly 5% during the Federal Reserve hiking period (2022-2024) as seen in figure 1[4]. Standard deterministic ML

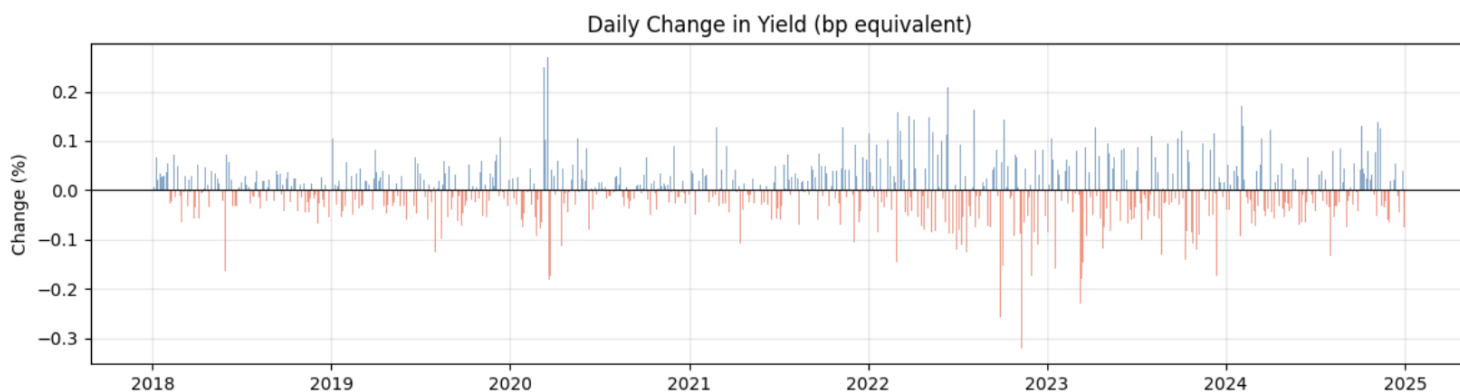
models would assume stable underlying distributions, making them poorly suited to this behaviour.

Figure 1 – US 10-Year Treasury Yield (1/1/2018 – 1/1/2025)



The daily yield changes also exhibit an excess kurtosis of 2.14. This confirms that a normal distribution is not suitable for the data and that the distribution of the yields has heavier tails than a Normal distribution[5]. A Normal distribution would suggest that it is extremely rare for a change of $\pm 0.25\%$ should occur, whereas in figure 2, it is observed this happens multiple times over the 7-year time period.

Figure 2 - Daily Change in Yield (Basis Points)



This proves that a student-t distribution would be better suited[5]. A deterministic machine learning model fit to the historical data would fail to predict and underestimate the probability of these heavy tail events, events that would influence the movements in ASX energy stock prices the most.

Additionally, the US bond market closes several hours before the ASX opens each morning, overnight yield movements are known at the time of the ASX open trade session. This structure means a Bayesian approach is necessary, due to the uncertainty in yield distribution propagating into next-day stock prices. Rather than producing a single point estimate, the Bayesian approach yields a full posterior distribution over model parameters.

Model Survey

Two broad approaches exist for quantifying uncertainty, Bayesian and Frequentist.

Frequentist methods treat model parameters as fixed but unknown constants observed from data. The most common method and the one used in this example is the Maximum Likelihood Estimation (MLE). MLE finds parameter values that maximise probability of the observed data, producing point estimates with confidence intervals.

Bayesian Markov Chain Monte Carlo (MCMC), implemented via PyMC's No-U-Turn Sampler (NUTS), samples the full posterior distribution over parameters, producing complete uncertainty estimates rather than single point values. Variational Inference (VI) was considered but given the moderate dataset size of 1761 observations and the 24-second acceptable sampling time, MCMC was preferred over VI.

Student-t likelihood was selected over Normal for both approaches due to the excess kurtosis of 2.14 and the rejection of normality by the Kolmogorov-Smirnov (KS) and Shapiro-Wilk (SW) p tests, 0.0122 and 0.000 respectively[5]. Maximum A Posteriori was also considered but discarded as it only returns the most probable posterior value, discarding full distributional uncertainty.

Human Factors and Ethics Review

When uncertainty is quantified, an honest picture of model confidence can be seen. This supports safer decision making and more informed tactics. It does not tell the user to “buy stock now” it gives a degree of for example “65% certain the stock will rise”, putting ownership back onto the user so they are aware of the risk. This negates the risk of false confidence in a model as well.

Conversely, a system relying solely on point estimates without uncertainty quantification risks projecting false confidence, leading to investors taking on excessive risk especially during previously mentioned hiking cycle.

This highlights the ethical obligation that lies in communicating uncertainty honestly and acknowledging model limitations.

Stakeholder Analysis

Two distinct stakeholders; a quantitative analyst and a financial regulator.

A financial regulator requires model transparency and explainability, probabilistic outputs must be interpretable and auditable to ensure compliance with financial regulations.

A Bayesian posterior distribution, whilst mathematically rigorous can be difficult to present in a transparent format for audit. These requirements directly contradict each other. The solution is to produce simplified uncertainty summaries such as confidence intervals and credible intervals alongside full posterior outputs.

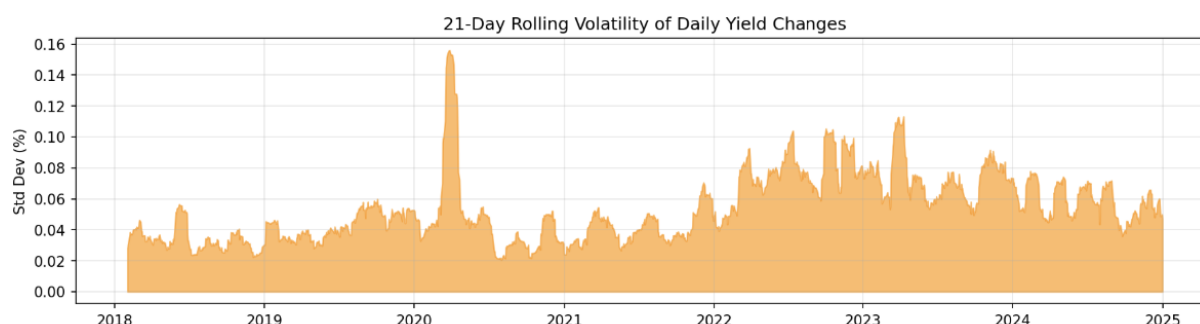
Dataset Analysis

Source

The data used in this study is the US 10-Year Treasury Yield sourced from Yahoo Finance (yfinance) via the ^TNX ticker[6]. The dataset spans 1761 trading days from 1st January 2018 to 1st of January 2025.

The dataset contains one feature, the closing yield expressed as a percentage. Difference from previous day close to current day close was applied to see the daily change. A 21-day rolling standard deviation was computed as a volatility proxy to capture time-varying uncertainty in the series, see figure 3.

Figure 3 - 21-Day Rolling Volatility of Daily Yield Changes



Exploratory Data Analysis

The dataset contains no missing values, US public holidays produce gaps in the data which are forward-filled to align with the ASX trading calendar. The exploratory analysis revealed 3 distinct macroeconomic regimes across the sample period, visible in figure 1. From 2018-2019 there are moderate yields with limited volatility, the near-zero yields during the COVID pandemic and the sharp hiking cycle post covid from 2022-2024.

The daily yield changes have a mean of 0.0012% and standard deviation of 0.0579%, with an excess kurtosis of 2.14 confirming heavy tailed behaviour. The 21-day rolling volatility displays clustering, importantly during the spike during the COVID pandemic and remaining elevated during the aforementioned hiking cycle, demonstrating that the variance parameter is itself time-varying and uncertain.

Frequentist and Bayesian Comparative Analysis

Figure 4 - Table of Comparative Results

Parameter	FREQUENTIST		BAYESIAN	
	MLE Estimate	MLE 95% Confidence Interval	Bayes Mean	Bayes 94% Credible Interval
μ (Mean Daily Change)	0.001198	[-0.001507, 0.003903]	0.001054	[-0.001332, 0.003477]
σ (Volatility)	0.057900	[0.055987, 0.059813]	0.046280	[0.043749, 0.048907]
ν (Degrees of Freedom)	5.27	N/A	5.52	[4.34, 7.14]

**Results may vary slightly due to the Monte Carlo method, as it is a stochastic sampling process results are subject to minor variations in runs. These changes are insignificant, with values changing by 0.000001-0.000005.

Both methods agree on the mean daily change, confirming a near-zero daily drift. The discrepancy in volatility arises from MLE inflating σ to compensate for heavy tails, whereas the Bayesian student-t model handles tail behaviour separately through the degrees of freedom ν . Both methods estimate ν similarly, 5.27 to 5.52, confirming heavy-tailed behaviour.

The most notable finding is the non-overlapping confidence and credible intervals for volatility. Due to the Frequentist Normal MLE having no dedicated tail parameter, volatility is inflated to compensate. The MLE confidence interval [0.055987, 0.059813] sits entirely above the Bayesian credible interval [0.043749, 0.048907]. This demonstrates that model choice has real consequences for the volatility signal fed into the DAG inference engine.

During Bayesian estimation, the four chains converged to the same posterior distribution with an R-hat of (μ 1.0002, σ 1.0000 and ν 1.0000), indicating the results are reliable.

System Architecture and Implementation

Design

Figure 5 - System Architecture Diagram

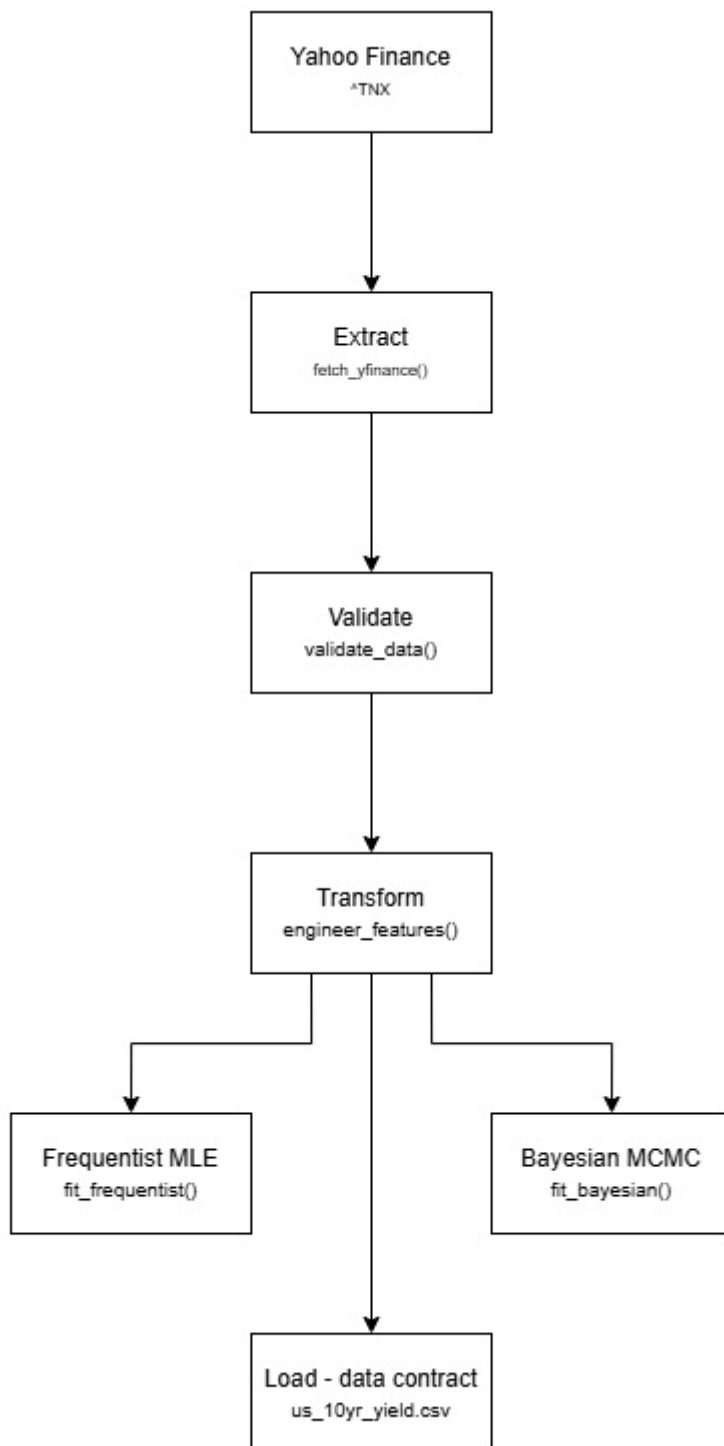


Figure 5 displays the architecture of the automated data ingestion and probabilistic estimation pipeline following an Extract, Transform, Load (ETL) structure. Data is extracted via Yahoo Finance, validated and transformed before flowing into the CSV data output. Frequentist and Bayesian estimation modules operate as side branches for analytical reasoning without affecting the main pipeline.

Process

Data has been extracted via yfinance using the ^TNX ticker. This dataframe is passed through the validate_data() function. This function returns the shape of the dataframe, any missing values, and a display of the date range. Validated data is then transformed via engineer_features(), computing daily yield changes and computing a 21-day rolling standard deviation as a volatility proxy. Transformed data is simultaneously passed to the Frequentist and Bayesian estimation modules as well as loaded directly to a csv.

Technical Decisions

For data extraction, yfinance was chosen as it requires no API key, it also is used by majority of group members allowing for easy integration for Artefact2. Pandas handled all data manipulation, scipy.stats provided MLE fitting for Normal and Student-t distributions and PyMC was selected for Bayesian Inference. The NUTS sampler was preferred over Gibbs sampling due to handling the continuous parameters more efficiently.

Appendix

Citations

- [1] A. Avdjiev, V. Bruno, C. Koch, and H. S. Shin, "The dollar exchange rate as a global risk factor: evidence from investment," *IMF Economic Review*, vol. 67, no. 1, pp. 151–173, 2019.
- [2] M. Neffelli and M. Resta, "Is VIX still the investor fear gauge? Evidence for the US and BRIC markets," *arXiv preprint*, arXiv:1806.07556, 2018. [Online]. Available: <https://arxiv.org/abs/1806.07556>. [Accessed: 15-May-2026].
- [3] "Impact of Geopolitical Risk and Crude Oil Prices on Stock Return," *Bulletin of Monetary Economics and Banking*, 2022. [Online]. Available: <https://bulletin.bmeb-bi.org/cgi/viewcontent.cgi?article=2158&context=bmeb>. [Accessed: 16-May-2026]
- [4] D. Bauer, B. Bernanke, and E. Milstein, "The Treasury Tantrum of 2023," *FEDS Notes*, Board of Governors of the Federal Reserve System, Sep. 2024. [Online]. Available: <https://www.federalreserve.gov/econres/notes/feds-notes/the-treasury-tantrum-of-2023-20240903.html>. [Accessed: 17-May-2026].
- [5] B. O. Bradley and M. S. Taqqu, "Financial Risk and Heavy Tails," in *Handbook of Heavy Tailed Distributions in Finance*, vol. 1, S. T. Rachev, Ed. Amsterdam, Netherlands: Elsevier, 2003, pp. 35–103
- [6] Yahoo Finance, "CBOE Interest Rate 10 Year T Note (^TNX)," *Yahoo Finance*, [Online]. Available: <https://finance.yahoo.com/quote/%5ETNX>. [Accessed: 15-May-2026].

AI Declaration

Claude (Anthropic, claude.ai, accessed May 2026) was used for idea generation and structural brainstorming only. All code, written analysis, and documentation was authored independently.

AI Model Used

Tool	Model	Usage Period
Claude.ai	Claude Sonnet 4.6 (Anthropic)	May 2026

Interaction Log Summary

#	Prompt Summary	AI Output Summary	How I Used It
1	Asked for a recommended feature for the individual component given time pressure and group allocations	AI recommended US 10-Year Treasury yield via ^TNX, citing data availability, causal story, and probabilistic behaviour	Used to confirm feature selection and understand justification arguments
2	Asked what the causal relationship would be between Treasury yield and ASX energy stocks	AI explained discount rate effect, USD strength proxy, and risk-free rate competition channels	Used as basis for causal argument in justification and team integration sections
3	Asked for explanation of Frequentist MLE and Bayesian MCMC from scratch	AI explained both methods, the difference between confidence intervals and credible intervals, and why σ differs between approaches	Used to deepen understanding before writing model survey and comparative analysis sections
4	Asked for explanation of heavy tails and kurtosis	AI explained leptokurtosis, Student-t degrees of freedom, and why Normal distribution underestimates tail risk	Used to understand and write the justification section around kurtosis finding of 2.14
5	Asked for help structuring the report	AI provided word budget breakdown and section-by-section structure mapped against rubric requirements	Used as a structural guide, all written content is my own
6	Asked what the MCMC convergence diagnostics (R-hat, ESS, mcse) mean	AI explained each column of the ArviZ summary output	Used to understand results before writing the comparative analysis section

#	Prompt Summary	AI Output Summary	How I Used It
7	Asked for IEEE citation formatting guidance	AI formatted citations for Yahoo Finance, PyMC, Bradley & Taqqu, and other sources	Used as formatting reference, all sources were independently verified
8	Asked why NUTS is more efficient than Gibbs sampling	AI explained gradient-based vs one-at-a-time sampling for continuous parameters	Used to understand and justify the technical decision in the report